The Universality of Electricity Demand: Zero-Shot Foundation Models Match Locally-Trained Accuracy on Brazil's Power Grid

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***Abstract*—Short-term load forecasting (STLF) has traditionally required models trained on years of local historical demand—a barrier in data-scarce grids. We ask whether recent time series *foundation models*, applied zero-shot, can forecast Brazilian electricity load with no local training. Using seven years (2019–2025) of hourly ONS load across all four subsystems of the interconnected grid (SIN), we benchmark Chronos-2 (120 M), TiRex (35 M), and Moirai 2.0 (11 M) against a hyperparameter-tuned N-BEATS, a linear model, an LSTM, and a seasonal-naive baseline. On the Southeast/Center-West (SE) subsystem at a 24-hour horizon, zero-shot Chronos-2 attains 1.86 % MAPE—statistically indistinguishable from a tuned N-BEATS trained on 5+ years of local data (1.91 % $\pm$ 0.08 %; Diebold–Mariano $p > 0.29$). Light fine-tuning yields only a 7 % relative gain (1.73 %). The result holds across all four subsystems (1.67–3.17 % MAPE, $R^2 > 0.90$), three test years (1.89 % $\pm$ 0.04 %), and—on the Tokyo (TEPCO) grid, a different country and hemisphere—where the same three zero-shot models again beat a locally-trained N-BEATS (3.91 % vs. 4.44 %). The dominant residual error concentrates on national holidays (4–5$\times$ the normal-day error); adding a single binary holiday covariate cuts holiday error by 35 %. We read this as evidence for a *bounded universality* of electricity demand: physics- and behavior-driven structure transfers across grids and hemispheres without adaptation, while calendar-specific demand does not.**

***Index Terms*—Short-term load forecasting, foundation models, zero-shot learning, power systems, Chronos, Moirai, TiRex, N-BEATS.**

I. Introduction

Short-term load forecasting (STLF) underpins unit commitment, economic dispatch, reserve sizing, and day-ahead market clearing. The dominant paradigm has each utility build a bespoke model trained on years of local load, augmented with weather and calendar features. This imposes a heavy barrier in data-scarce contexts—newly interconnected regions, microgrids, and developing grids lacking long, clean records.

Time series foundation models (TSFMs)—large networks pretrained on billions of observations across diverse domains—raise a sharp question: can a model that has never seen a particular grid's data forecast its load accurately? If so, forecasting could be deployed without the data-collection and maintenance overhead of local training.

We frame this as a hypothesis of *universality*. Demand is shaped by two kinds of forces. The first are universal—the physics of heating, cooling, and lighting; the circadian rhythm of activity; the weekly work/rest cycle—producing daily peaks, overnight troughs, and weekend dips in every electrified society. The second are local—national holidays, hydro-dependent supply constraints, and southern-hemisphere seasonality inverted from the northern-hemisphere data these models consume. If a pretrained model has internalized the universal component, it should transfer to an unseen grid zero-shot, with residual error confined to the local component.

We test this primarily on the Brazilian interconnected system (SIN) using public ONS data, with a cross-country replication on the Tokyo (TEPCO) grid. Our contributions are: (1) a systematic benchmark of 2024-era TSFMs on Brazilian load (all four SIN subsystems, four horizons, three test years) plus a cross-country replication on a Japanese grid; (2) evidence that zero-shot models match or beat a tuned, locally-trained N-BEATS (Diebold–Mariano $p > 0.29$ on SE); (3) a precise account of *when and why* they fail—and a holiday covariate that recovers most of the gap; and (4) a reproducible benchmark.

II. Related Work

Brazilian STLF work relies on locally-trained SVR, ANN, ARIMA, and LSTM models on ONS or utility data [6], [7], [8], all requiring per-grid development. The TSFM paradigm emerged in 2023–2024—pretrain on billions of time points, then forecast unseen series zero-shot—with models including Chronos [1], TimesFM [3], Moirai [4], and TiRex [5]. TSFMs have been evaluated on US grids (ERCOT [9]), Singapore/Australia [10], and European households [11], but not on emerging-market grids with hydro dependency, inverted seasonality, and distinct holiday calendars, nor against a locally-trained deep-learning baseline on the same data. We fill that gap.

III. Data and Methodology

Data. Hourly load from the ONS open-data portal (dados.ons.org.br, CC-BY 4.0) for the four SIN subsystems

TABLE I
SIN SUBSYSTEMS AND DATASET STATISTICS (2019–2025)

Sub.	Region	Mean (MW)	Test (MW)	Share
SE	SP, RJ, MG	40,408	44,226	∼55 %
S	PR, SC, RS	12,320	13,804	∼17 %
NE	BA, PE, CE	11,711	13,267	∼17 %
N	AM, PA	6,641	8,312	∼11 %

(Table I); SE is the largest at ∼55 % of national load (which averages 71,081 MW). Each series spans 2019-01-01 to 2025-12-31 (61,368 hourly rows). The most recent 365 days (8,760 h) form the test set: foundation models see only the preceding context window (zero-shot, never trained on ONS data); trained baselines use all earlier data, reserving the prior 60 days for validation.

Task. Given a context window of hourly load, predict the next 24 h (day-ahead); we also evaluate 168/336/720 h. Forecasts roll through the test year in 24-h steps and are averaged.

Models. Zero-shot: *Chronos-2* (120 M, encoder-only, quantile output; median reported), *TiRex* (35 M, xLSTM), *Moirai 2.0* (11 M). We verified ONS data is *absent* from Chronos's pretraining corpus [1], [2] (no data leakage). Trained: *N-BEATS* (7.3 M, Darts; primary baseline), *Linear* (∼8 K), *LSTM*, and *Naive (7-day-ago)*. Metrics: MAE, RMSE (MW), MAPE, MASE/RMSSE (seasonality = 24), R^2; CRPS and prediction intervals for probabilistic eval; Diebold–Mariano (DM) test on per-window errors. Hardware: Mac Mini M4, no GPU; MPS fails for Darts N-BEATS and Chronos fine-tuning, run on CPU.

IV. RESULTS

A. Main Comparison (SE, 24 h)

Table II and Fig. 1 report the headline. Zero-shot Chronos-2 achieves **1.86 % MAPE**, edging out a tuned N-BEATS (1.91 %) trained on five years of local data, with Moirai close behind (1.93 %); all TSFMs beat naive (5.13 %) by a wide margin. The DM test gives $p > 0.29$ against all three N-BEATS seeds and bootstrap 95 % CIs overlap (Chronos-2 CI $[1.70\%, 2.04\%]$): Chronos-2 vs. N-BEATS is a *statistical tie*—but Chronos-2 needs *zero* local training. Light fine-tuning (400 steps, lr 1e-5, ∼40 min CPU) reaches 1.73 % (a 7 % gain), showing pretraining already captures the dominant patterns. The LSTM failed to train competitively and is shown for completeness. For context, Chronos-2 (load only) rivals proprietary ISO systems using weather/calendar inputs: PJM 24 h (1.78–1.98 %), ERCOT (1.66–3.73 %).

N-BEATS tuning. To ensure fairness, N-BEATS was grid-searched over input length $\in \{168, 336, 720\}$ h and lr $\in$ {1e-4, 5e-4, 1e-3}. The best config (168 h, lr 5e-4), re-run with seeds $\{42, 123, 7\}$, gives 1.85/1.88/2.00 % $\rightarrow$ **1.91 % ± 0.08 %**—a 0.23 pp improvement over its untuned setting (2.14 %), confirming the baseline is properly optimized.

TABLE II
MAIN RESULTS — SE, 24 H HORIZON

Model	Type	MAPE %	MASE	R^2
Chronos-2	**Fine-tuned**	**1.73**	0.30	0.96
Chronos-2	Zero-shot	1.86	0.33	0.96
N-BEATS (tuned)	Trained 5 yr	1.91 ± 0.08	0.34	0.96
Moirai 2.0	Zero-shot	1.93	0.34	0.95
Linear	Trained	2.26	0.40	0.94
TiRex	Zero-shot	2.33	0.40	0.94
Naive (7 d)	Baseline	5.13	0.89	0.77
LSTM	Trained	13.14	2.39	−0.36

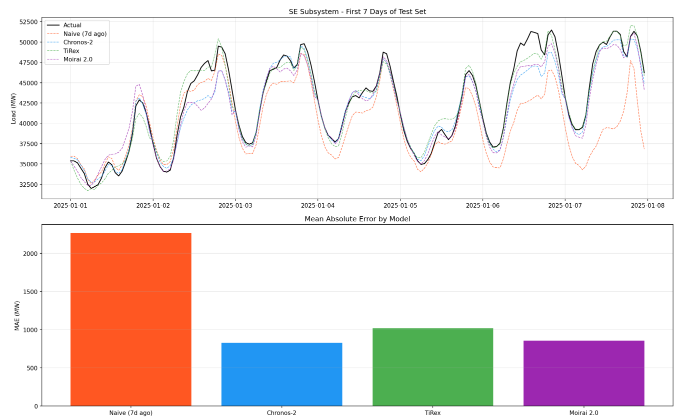

Fig. 1. SE, 24 h: 7-day forecast comparison and per-model MAE. Zero-shot Chronos-2 tracks the load curve closely; the naive baseline lags on the daily ramp.

B. Generalization

Across subsystems. Chronos-2 leads on all four (Table III), 45–64 % above naive, $R^2 > 0.90$ everywhere. The smallest subsystem (N) is easiest (1.67 %); S is hardest (3.17 %), likely from winter cold-front heating. On S, tuned N-BEATS reaches 3.26 % ± 0.21 % vs. 3.17 %—again a tie.

Horizon. Accuracy degrades with horizon, with a *naive crossover* at ∼2–3 weeks. SE Chronos-2 MAPE: 1.86 % (24 h), 3.59 % (168 h), 4.18 % (336 h), 5.69 % (720 h); at 720 h Chronos-2 (MASE 1.02) and Moirai (1.03) fall behind naive while only TiRex (0.91) stays ahead. TSFMs are thus best for operational (24–168 h) horizons.

Multi-year. Chronos-2 is stable across test years 2023/2024/2025 (1.94/1.87/1.86 %, mean 1.89 % ± 0.04 %) while naive varies more (5.31 % ± 0.17 %).

Context length. The critical threshold is *one week*: MAPE halves (5.65 %→2.80 %) once a full weekly cycle is seen, reaching 1.86 % at 30 days; beyond that, returns are marginal (90 days: 1.77 %). Thirty days is the practical accuracy/compute sweet spot.

Model scale. The Chronos-Bolt family scales log-linearly (Fig. 2): Tiny 9 M $\rightarrow$ 2.28 %, Mini 21 M $\rightarrow$ 2.19 %, Small 48 M $\rightarrow$ 2.02 %, Chronos-2 120 M $\rightarrow$ 1.86 %. At matched size zero-shot Bolt-Tiny (2.28 %) loses to tuned N-BEATS (1.91 %), but Moirai (11 M, 1.93 %) nearly ties it—architecture matters as much as scale.

TABLE III
MAPE (%) BY SUBSYSTEM — 24 H

Sub.	Naive	Chronos-2	TiRex	Moirai
SE	5.13	**1.86**	2.33	1.93
S	7.11	**3.17**	3.37	3.35
NE	3.76	**1.94**	2.06	2.05
N	3.03	**1.67**	1.67	1.76

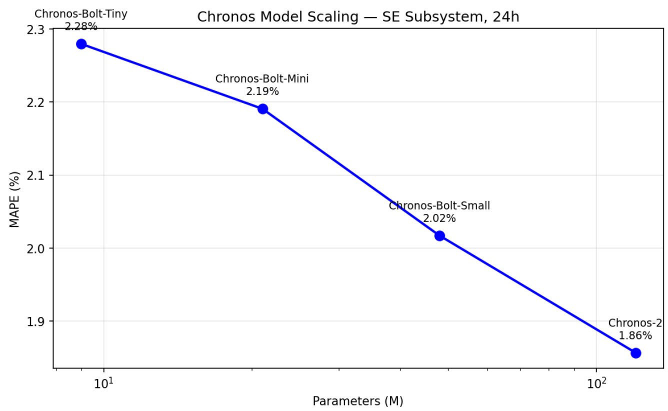

Fig. 2. Chronos scaling: MAPE decreases log-linearly with parameter count; at ∼9 M zero-shot trails tuned N-BEATS, and by 120 M it closes the gap.

Probabilistic. Both probabilistic models are well-calibrated but slightly conservative—80 % PIs cover 86–89 % of outcomes, desirable for reserve scheduling. Chronos-2 has the better CRPS (643 vs. 688 MW) and Winkler score (4354 vs. 4378); Moirai produces sharper intervals (3024 vs. 3295 MW) despite being 11× smaller.

C. Cross-Country Replication (Tokyo)

To probe universality beyond Brazil, we repeat the benchmark on the TEPCO grid (Tokyo, ∼32 GW, 2019–2024), with N-BEATS trained on 2019–2023 Japanese data and 2024 held out (Table IV). All three zero-shot foundation models again *beat* the locally-trained N-BEATS, in the identical ranking observed for Brazil (Chronos-2 $>$ Moirai $>$ TiRex $\gg$ naive). Absolute MAPE is higher (Japanese demand is more volatile), but the qualitative result—zero-shot transfer matches or exceeds dedicated local training—holds across a different country, hemisphere, and holiday calendar. This is the strongest single piece of evidence for the universality hypothesis.

V. Error Analysis: When and Why

Error follows the load curve (Fig. 3): lowest in the overnight trough (0.51 % at hour 0), highest mid-afternoon (2.72 % at hour 15). Weekends (1.75 %) beat weekdays (1.90 %); Monday is hardest (2.37 %), Tuesday easiest (1.60 %).

The central finding is the holiday regime (Table V). On Brazilian public holidays Chronos-2 degrades to several times its normal-day error, because nothing in the global corpus signals a Brazilian holiday, so it forecasts ordinary workday demand. The effect is a property of the data, not one model: on the same holidays Moirai reaches 7.9 % and TiRex 9.1 % MAPE, and the naive baseline 12.8 %, against ∼1.6 % on normal days.

TABLE IV
CROSS-COUNTRY: TEPCO TOKYO — 24 H, TEST 2024

Model	Type	MAPE %	MASE	R^2
Chronos-2	Zero-shot	**3.91**	0.59	0.91
Moirai 2.0	Zero-shot	3.94	0.59	0.91
TiRex	Zero-shot	4.05	0.61	0.91
N-BEATS	Trained 5 yr	4.44	0.67	0.89
Naive (7 d)	Baseline	8.86	1.30	0.65

TABLE V
HOLIDAY COVARIATE EXPERIMENT — CHRONOS-2, SE (TEST 2024)

Regime	Hours	Zero-shot	+Cov.	Δ %
Normal	5688	1.53	1.53	+0.0
Weekend	2232	1.77	1.74	−2.0
Day-before	264	2.28	2.19	−4.2
Day-after	264	2.29	1.93	−15.8
Carnaval	48	5.68	4.64	−18.3
Holiday	264	**6.76**	**4.37**	**−35.4**
Bridge	24	3.02	6.12	+102.5
Overall	8784	1.82	1.73	−5.0

This is precisely where a small dose of local knowledge should help—and it does. Adding a single binary holiday covariate and lightly fine-tuning Chronos-2 (test 2024) cuts holiday MAPE by 35 % (6.76 % → 4.37 %), Carnaval by 18 %, and the day after a holiday by 16 %, while normal days are untouched and overall MAPE falls 5 % (Table V). The lone regression—"bridge" days between a holiday and a weekend worsen (3.02 % → 6.12 %), on only 24 test hours—marks where the coarse binary encoding is too blunt and a richer calendar feature is needed. This is exactly the bounded-universality prediction: the residual error is local calendar knowledge, and supplying that knowledge recovers most of the gap.

The same covariate hurts a trained model. A natural control: does the holiday flag help the locally-trained N-BEATS as much? We add the identical feature to the tuned N-BEATS (3 seeds). It does the *opposite*—overall MAPE worsens from 1.91 % to 2.20 % (+15 %) and holiday MAPE from 6.00 % to 6.30 % (+5 %). The interpretation sharpens the universality boundary: N-BEATS, trained on 5+ years of local data, has *already* absorbed Brazilian holidays implicitly into its weights, so an explicit flag is redundant and merely adds noise; the zero-shot model never saw those holidays and gains from being told. The same intervention helping the foundation model and hurting the trained one is direct evidence that what the foundation model lacks—and the flag supplies—is precisely *local calendar knowledge*.

VI. Discussion and Conclusion

The results support a universality of electricity demand *with a precise boundary*. Components driven by shared physical and behavioral forces—diurnal cycles, the workweek,

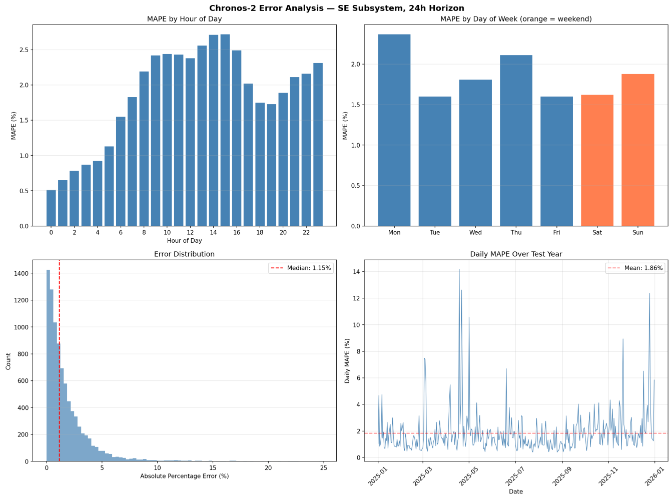

Fig. 3. Error decomposition: MAPE by hour-of-day and day-of-week, error distribution, and daily MAPE over the test year (spikes coincide with holidays).

seasonality—transfer to an unseen grid well enough that a zero-shot model ties or beats a locally-trained one across subsystems, years, and—on the Tokyo grid—a second country and hemisphere. Components driven by local culture—national holidays—do not, degrading 4–5×; but supplying exactly that local knowledge via a binary holiday covariate recovers 35 % of the holiday gap, confirming the boundary is calendar knowledge rather than an architectural limit.

This yields an operational recipe: (1) start zero-shot (Chronos-2, or the 11 M Moirai when resource-constrained) with ∼30 days of context; (2) fine-tune if resources permit (∼7 % gain, 40 min CPU); (3) add a binary holiday covariate (cut holiday error 35 % in our experiment); (4) do not use beyond ∼2 weeks, where naive wins; (5) one week of history is the minimum viable context. *Limitations:* univariate input (no weather); two countries (Brazil, Japan) remain a small sample of grids; load-only trained baselines, so a weather-augmented TFT could narrow the gap; the holiday covariate's binary encoding regresses on rare "bridge" days; and the LSTM did not converge competitively.

In sum, a model pretrained on global time series, with no exposure to the target grid's data, matches a tuned model trained on 5+ years of local Brazilian data (1.86 % vs. 1.91 % ± 0.08 %; DM $p > 0.29$) across all four subsystems and three years, and *beats* a locally-trained N-BEATS on the Tokyo grid (3.91 % vs. 4.44 %). The boundary of universality is precise—the model fails on culturally-local holidays and essentially nowhere else—and a single holiday covariate largely closes even that gap. For operators in data-scarce grids, accurate day-ahead forecasting no longer requires years of local model development: thirty days of load history and an off-the-shelf foundation model suffice.

Acknowledgment

The author thanks the course instructor for suggesting cross-market generalization (an Asian-market extension is in progress) and reviewers for the N-BEATS tuning and data-leakage audits incorporated above.

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